Criticality of the Hopf bifurcations in the Becker–Döring equations with atomization

chemistry-open-problems · 18 October 2026 · partial result, independently verified (confirmed with corrections)

Solver: Claude Opus 5.5 (effort xhigh), sub-agent of the orchestrator; citing-works search by a nested Claude Opus 5.5 sub-agent (effort xhigh) · LaTeX edition: Claude Opus 5.5 (effort xhigh) · Independent verification and corrections: Claude Opus 5.5 (effort xhigh), fresh context, no access to the solver's reasoning

Abstract. Pego and Velázquez proved that the Becker–Döring equations with atomization undergo Andronov–Hopf bifurcations at $\kappa = \kappa_j(M)$ for large maximal cluster size M, and wrote that they had not managed to determine analytically whether the bifurcating periodic solutions are stable. We evaluate the first Lyapunov coefficient ℓ_1 in exact closed form at a cost independent of M, enclose the Hopf points by a Krawczyk operator in ball arithmetic, and certify the sign of ℓ_1 for every integer M in $15 \le M \le 10^5$ on the first branch ($44 \le M \le 10^5$ and $87 \le M \le 10^5$ on the second and third): $\ell_1 < 0$ (supercritical) except at $M = 15$ on the first branch, where $\ell_1 > 0$ (subcritical). The remaining spectral hypothesis of the Hopf theorem is certified for $M \le 160$. The first-branch periodic orbits are therefore rigorously orbitally stable for $16 \le M \le 160$. For larger M this conclusion rests on that hypothesis, which was checked only in floating point. An independent verification re-proved the signs for $M \le 300$ with different coordinates and code. It also showed by exact algebra that, for $15 \le M \le 57$, $\kappa_1(M)$ is the first loss of stability. A formal reduction for $M \to \infty$ to a scalar renewal equation predicts a negative Landau coefficient on every branch, and certified values up to $M = 10^8$ approach the predicted limit. The infinite range $M > 10^5$ remains unproved.

1 Problem and source

R. L. Pego, J. J. L. Velázquez, *Temporal oscillations in Becker–Döring equations with atomization*, Nonlinearity 33 (2020) 1812–1846, doi:10.1088/1361-6544/ab6815, arXiv:1905.02605 (v2 read in full, LaTeX source).

Theorem 3.3 of the paper states that for $k \in \mathbb{N}$ and $M > M_{0,k}$ the system admits a Hopf bifurcation as $\kappa = K\sqrt{M}$ passes through each $\kappa_j(M)$, $j = 1, \dots, k$. Immediately after it the authors write (verbatim, §3.3):

> "We have not managed to determine analytically whether the bifurcating solutions are stable (the supercritical case) or not. Many of our numerical computations, as in Fig. 1, are consistent with the presence of stable periodic solutions, however."

The constant $M_{0,k}$ is not explicit.

Novelty check (2026-10-18): 20 citing works (OpenAlex and Semantic Scholar) were checked; every arXiv full text was grepped. Later work by the authors and searches on key phrases were also checked. Nothing settles the question. The closest work is Niethammer–Pego–Schlichting–Velázquez (SIAM J. Appl. Math. 2022, arXiv:2102.06751). It carries out a *formal* direction-of-bifurcation computation for a *different* model (Becker–Döring

with injection and depletion). Details are in `references.md`.

2 Model, linearization, exact quadratic structure

The unknowns are $n_\ell \geq 0$, $1 \leq \ell \leq M$, with $N = M - 1$ and $J_\ell = n_\ell n_1 - n_{\ell+1}$ for $1 \leq \ell \leq N$ (paper (1)–(4)):

$$\dot{n}_\ell = J_{\ell-1} - J_\ell \ (2 \leq \ell \leq N), \qquad \dot{n}_M = J_N - K n_M, \qquad \dot{n}_1 = -J_1 - \sum_{\ell=1}^{N} J_\ell + MK n_M.$$

The mass $\sum_\ell \ell n_\ell$ is conserved.

The constant state $n_\ell \equiv A := 1 + K$ is an equilibrium. Write $n = A\mathbb{1} + v$. A direct expansion (checked in `scripts/01_check_model.py`, claims M1–M3) gives the **exact** identity

$$\dot{v} = Bv + v_1\, Sv, \tag{2.1}$$

with no cubic or higher terms. Here B is the paper's matrix (27). The linear map S is

$$(Sv)_1 = -2v_1 - \sum_{\ell=2}^{N} v_\ell, \qquad (Sv)_\ell = v_{\ell-1} - v_\ell \ (2 \leq \ell \leq N), \qquad (Sv)_M = v_N.$$

Equivalently, $f(n) = (n_1 - 1)\, Sn + Tn$ with T linear. The nonlinearity is the product of the "transport speed" $c = n_1 - 1$ and a fixed linear operator.

Mass conservation: $\boldsymbol{\ell} = (1, 2, \ldots, M)$ satisfies $\boldsymbol{\ell} B = 0$ and $\boldsymbol{\ell} S = 0$. Hence the hyperplane $H = \{\boldsymbol{\ell} \cdot v = 0\}$ is invariant under (2.1), and $\operatorname{Ran} B = H$.

By paper Lemma 3.1, 0 is a simple eigenvalue of B, with eigenvector $\bar{v} \notin H$. Hence $B|_H : H \to H$ is invertible.

Kuznetsov's multilinear forms for (2.1) are

$$\mathcal{B}(x, y) = x_1 Sy + y_1 Sx, \qquad \mathcal{C} \equiv 0.$$

3 Hopf setting and first Lyapunov coefficient

Setting. As in the paper, the bifurcation parameter is κ, and the dynamics is restricted to the invariant affine hyperplane through $A\mathbb{1}$. In the coordinate v this is H, a fixed space, and the equilibrium is $v = 0$ for every κ.

Eigenvalues. The eigenvalues of $B|_H$ are those of B except the simple eigenvalue 0. At a Hopf point κ_j, B has eigenvalues $\pm i\omega$ with $\omega > 0$, and $\operatorname{Re} d\lambda/d\kappa > 0$ (paper (33), re-certified in §5).

Coefficient. Let $Bq = i\omega q$, let u be a row vector with $uB = i\omega u$, normalized by $uq = 1$, and set $\langle p, x \rangle := ux$. Kuznetsov's formula (*Elements of Applied Bifurcation Theory*, 3rd ed., §3.5), with $\mathcal{C} = 0$, reads

$$\ell_1 = \frac{1}{2\omega} \operatorname{Re}\Big[-2\, u\, \mathcal{B}(q, w_{11}) + u\, \mathcal{B}(\bar{q}, w_{20}) \Big],$$

$$w_{11} = (B|_H)^{-1} \mathcal{B}(q, \bar{q}) \in H, \qquad w_{20} = (2i\omega - B)^{-1} \mathcal{B}(q, q). \tag{3.1}$$

Why the full-space objects may be used.

- $q \in H$, because $\boldsymbol{\ell} q = \boldsymbol{\ell} B q/(i\omega) = 0$.
- $\mathcal{B}(x, y) \in H$ for all x, y, because $\boldsymbol{\ell} S = 0$.
- $u|_H$ is the adjoint eigenvector of $B|_H$.
- $w_{20} \in H$ automatically, because H is invariant under B and $2i\omega \notin \operatorname{spec} B$.
- w_{11} must be taken as the unique solution of $Bw = \mathcal{B}(q, \bar{q})$ **in** $\boldsymbol{H}$. Adding a multiple of $\bar{v}$ would change ℓ_1.

Formula (3.1) is basis-free (projection method), so it is the first Lyapunov coefficient of the restricted system.

Sign convention. ℓ_1 depends on the normalization of q only through a positive factor. Two normalizations are used: $q^H q = 1$ ("unit") and $q_1 = 1$.

Since $\operatorname{Re} \lambda'(\kappa_j) > 0$:

- $\ell_1 < 0$ means **supercritical**: periodic orbits exist for $\kappa > \kappa_j$, $\kappa \approx \kappa_j$, with amplitude $\propto (\kappa - \kappa_j)^{1/2}$, and they attract within the centre manifold.
- $\ell_1 > 0$ means **subcritical**.

For $j = 1$, when all other eigenvalues of $B|_H$ have negative real part (hypothesis H2 and the count of §5), a supercritical orbit is orbitally asymptotically stable within H. For $j \geq 2$ the equilibrium already has $j - 1$ unstable pairs, so the bifurcating orbits are unstable in the full system whatever the sign. The criticality is then a centre-manifold statement.

Remark (intrinsic). ℓ_1 depends only on the vector field at the critical point. The conclusion therefore holds for any one-parameter path through the Hopf point in the (K, mass) plane that crosses transversally. An example is K varying at fixed total mass, where the equilibria are the non-constant ones of paper (15).

4 Exact closed-form evaluation of $\boldsymbol{\ell_1}$ for any M

All vectors occurring in (3.1) have the form

$$v_\ell = \sum_e c_e \, \beta_e^\ell + \sum_{k \in P} d_k \, \delta_{\ell k}, \qquad P \subset \{1, 2, N, M\}, \tag{4.1}$$

with finitely many bases β_e. Each base is a monomial $\varphi^{e_0} \bar{\varphi}^{e_1} A^{e_2} s^{e_3} \bar{s}^{e_4}$ with integer exponents; φ and s are defined below. All operations on such vectors are exact and cost $O(1)$ in M. Implementation: `scripts/bdlib.py`, classes `Problem` and `Seq`.

Right eigenvector (paper §5). For $\lambda = (A - \varphi^{-1})(\varphi - 1)$ and $\psi = 1/(A\varphi)$:

$$q_\ell = c_1 \varphi^{M-\ell} + c_2 \psi^{M-\ell}, \qquad (c_1, c_2) = (f(\psi), -f(\varphi)), \qquad f(\varphi) = A\varphi^{M-1} + 1 - \varphi^{-1},$$

valid for $1 \leq \ell \leq M$ whenever φ is a root of the paper's F (eqs. (71), (76)).

Left eigenvector. The transposed equations, columns $2 \leq k \leq N$ of $uB = \lambda u$, read

$$u_{k-1} - (A + 1 + \lambda) u_k + A u_{k+1} = K u_1.$$

Hence $u_k = a_1 \varphi^k + a_2 \psi^k + u_c$, with $u_c = -K u_1/\lambda$ and $u_1 = (a_1 \varphi + a_2 \psi)\lambda/(\lambda + K)$. Columns 1 and M give a singular 2×2 system for (a_1, a_2); one row of it fixes the null vector.

Linear maps. B and S act on a term β^ℓ as multiplication by $A/\beta-(A+1)+\beta$ (bulk rows of B) or $1/\beta-1$ (bulk rows of S). Rows 1 and M, and the neighbours of point corrections, are evaluated exactly. The full sums in row 1 are geometric sums. The differences are stored as point corrections.

Solving $(\boldsymbol{B-\mu})\boldsymbol{w}=\boldsymbol{r}$ for r of the form (4.1) (here $\mu=0$ or $2i\omega$). Ansatz:

$$w=\sum_e \frac{c_e}{A/\beta_e-(A+1)+\beta_e-\mu}\,\beta_e^\ell + h_1 s_1^\ell + h_2 s_2^\ell + e_1\delta_{\ell 1} + e_M\delta_{\ell M},$$

where $s_{1,2}$ are the roots of $s^2-(A+1+\mu)s+A=0$:

- for $\mu=0$: $s\in\{1,A\}$;
- for $\mu=2i\omega$: one new generator s, the other root being A/s.

The bulk rows $3\le\ell\le N-1$ hold identically. Rows $2, N$ (which involve w_1, w_M) and rows $1, M$ give four linear equations for (h_1,h_2,e_1,e_M).

Why this family contains the true solution. Rows $2,\dots,N$ form a three-term recurrence. Their solution set is an affine space of dimension 2, and it is the 2-dimensional subfamily of the ansatz fixed by rows 2 and N.

The case $\mu=0$. Row 1 is redundant, since $\boldsymbol{\ell}\,(Bw-r)=0$ whenever rows $2,\dots,M$ hold. It is replaced by the condition $\sum_\ell \ell w_\ell=0$, which selects $w_{11}\in H$. The 4×4 system is solved in ball arithmetic (`acb_mat.solve`). Success proves that it is nonsingular, so the computed ball contains the exact solution. The resulting point corrections are numerically zero ($\sim 10^{-35}$), as they should be when r has no correction at rows 2 and N.

Inner products. $\sum_{\ell=1}^{M} u_\ell v_\ell$ is computed with exact geometric sums $\sum\beta^\ell=(\beta^{M+1}-\beta)/(\beta-1)$. A base that is identically 1 (zero exponent vector, e.g. $\psi\cdot A\varphi$) contributes the exact count M. Products β^ℓ of size $A^{\pm M}$ are harmless in Arb's arbitrary-exponent balls.

Cross-checks.

- The closed form agrees with a dense floating-point implementation of (3.1) to 10^{-12}–10^{-14} relative error for $M=20,\dots,200$, $j\le 4$ (`01_check_model.py`, M6).
- The eigenvector residuals are at rounding level.

5 Certification

Hopf points. Unknowns $\kappa=K\sqrt{M}$ and $z=M(\varphi-1)$ (the paper's scaling). The system is

$$Q^\varepsilon(z;\kappa):=K^{-3}F(1+z/M)=0\ \text{(complex)},\qquad \operatorname{Re}\Lambda=0,\quad \Lambda=M\lambda/K.$$

This is 3 real equations in 3 real unknowns, with $F=f_1g_2-g_1f_2$ exactly as in the paper, eqs. (71)–(76).

- Derivatives come from forward-mode dual numbers over Arb balls.
- The Krawczyk operator $\mathcal{K}(X)=m-Yf(m)+(I-YJ(X))(X-m)$ is evaluated on a box X of relative radius 10^{-25}. The inclusion $\mathcal{K}(X)\subset\operatorname{int}X$ proves existence and uniqueness of a zero in X.

- By paper Prop. 5.3 a zero is an eigenvalue $\lambda = (A - \varphi^{-1})(\varphi - 1)$ of B, which is purely imaginary at the zero ($\operatorname{Re}\Lambda = 0$). It is not spurious, since $\operatorname{Im}\varphi \neq 0$.

Nondegeneracy (certified at every point).

- *Simplicity:* $0 \notin \partial_z Q^\varepsilon$ on the box. Then φ is a simple root of F, and by paper Lemma 5.4 $i\omega$ is a simple eigenvalue of B.
- *Transversality:* $\operatorname{Re}(\Lambda + \kappa\, d\Lambda/d\kappa) > 0$ along the root curve, with $dz/d\kappa = -\partial_\kappa Q^\varepsilon / \partial_z Q^\varepsilon$. This equals $M^{3/2} \operatorname{Re} d\lambda/d\kappa$.

Sign of ℓ_1. The closed form of §4 is evaluated with $K = \kappa/\sqrt{M}$ and $\varphi = 1 + z/M$ given as the Krawczyk enclosure. The formulas are exact identities at the true Hopf point, so the output ball contains the true ℓ_1. Precision is 192 bits. The Krawczyk boxes have relative radius 10^{-25}, and the resulting ℓ_1 balls relative radius 10^{-36}–10^{-29} (measured at $M = 15$ and $M = 10^5$). The sign is therefore never in doubt.

Remaining hypothesis H2 (no other eigenvalue of $B|_H$ on $i\mathbb{R}$).

- For $M \leq 120$ it is certified with Arb's rigorous eigensolver (`acb_mat.eig`, `07_spectrum_H2.py`). All M eigenvalues of $B(K)$, with K the certified ball, are enclosed and proved simple. Exactly one enclosure contains 0, exactly two meet $i\mathbb{R}$ (at $\pm i\omega$), $2(j-1)$ have positive real part and the rest negative real part.
- This also shows rigorously that exactly $j-1$ conjugate pairs are unstable at $\kappa_j(M)$, so the crossing is the j-th one counted at that parameter value. It does not by itself exclude pairs that cross and re-cross the axis at smaller κ; see the remark on the word "first" in §6.
- Result: all 217 points with $M \leq 120$ PASS (106 for $j = 1$, 77 for $j = 2$, 34 for $j = 3$); log in `certificates/logs/07_spectrum_H2.log`. The verifier's independent code extends this certificate to $M \leq 160$ on all three branches (§11).
- For larger M, H2 and the branch label j are supported by floating-point eigenvalue counts: at $\kappa_j(1 \mp 10^{-7})$, $j-1$ resp. j unstable pairs, for all $M \leq 300$ and every 50th $M \leq 2000$. For $M > 2000$ there is no direct check of H2 or of the label. The label rests on the continuation in M: $\kappa_j(M)$ is strictly decreasing and $\kappa_j(M)$ and $\operatorname{Im} z_j(M)$ vary smoothly along each certificate file. The paper's Theorem 3.2 gives H2 for $M > M_{0,k}$, but $M_{0,k}$ is not explicit.
- *Simplicity and the paper's Lemma 5.4.* The first step of the proof of Lemma 5.4 (the 2×2 matrix $\mathcal{D}(\varphi)$ is not zero) uses a convexity argument that is valid only for real φ. At the certified points φ is not real. The step is then replaced by a direct check: $f(\varphi) \neq 0$ or $f(1/(A\varphi)) \neq 0$. This check is implicit in the certificate, because the eigenvector q, with coefficients $(c_1, c_2) = (f(\psi), -f(\varphi))$, is normalised by a nonzero ball. The verifier also checked it explicitly on a sample (§11). The rest of the proof of Lemma 5.4 then applies verbatim.

Generation and independent re-check.

- The sweep files are generated by `03_certify_range.py` (continuation in M).
- `04_verify_certificates.py` then re-proves every line from the stored midpoints alone, without continuation: Krawczyk, ℓ_1 enclosure, simplicity, transversality. It also checks that every integer M of the range is present.
- Done for every line of all six files ($3 \times (\approx 10^4 + 9 \cdot 10^4)$ Hopf points). Logs are in `certificates/logs/04_verify_*.log`.

6 Results on the finite range

Theorem A (computer-assisted; certificates in `certificates/`). For every pair (M, j) in

$$\mathcal{S} = \{(M,1) : 15 \le M \le 10^5\} \cup \{(M,2) : 44 \le M \le 10^5\} \cup \{(M,3) : 87 \le M \le 10^5\}$$

the scaled Hopf system has a unique zero $(\kappa_j(M), z_j(M))$ in an explicit box. There:

- $\pm i\omega$ is a simple pair of eigenvalues of B;
- $\operatorname{Re} d\lambda/d\kappa > 0$;
- the first Lyapunov coefficient of the system restricted to the mass hyperplane satisfies

$$\boxed{\ell_1 < 0 \ \text{ for all } (M,j) \in \mathcal{S}\backslash\{(15,1)\}, \qquad \ell_1 > 0 \ \text{ for } (M,j) = (15,1).}$$

Consequences (wording corrected at verification):

- Wherever the remaining hypothesis H2 holds, the Hopf bifurcation at $\kappa_j(M)$ is **supercritical**, except for $M = 15$ where it is **subcritical**.
- *Rigorous range.* H2 is certified for every $M \le 160$ on all three branches (solver: $M \le 120$; verifier, independent code: $M \le 160$, and also $M = 250, 300$). Hence, for every integer $16 \le M \le 160$, the Hopf bifurcation at $\kappa_1(M)$ is supercritical and its periodic orbits are orbitally asymptotically stable within the mass hyperplane. At $M = 15$ it is subcritical.
- *Conditional range.* For $160 < M \le 10^5$ the same conclusion holds *provided* H2 holds. H2 is checked only in floating point, for $M \le 300$ and every 50th $M \le 2000$. It is not checked at all for $M > 2000$. The paper's Theorem 3.2 gives H2 for $M > M_{0,k}$, but $M_{0,k}$ is not explicit, so it may lie beyond 10^5. The sign $\ell_1 < 0$ itself is certified on the whole range.
- *The word "first".* The stability of the bifurcating orbits only requires the spectrum at $\kappa_1(M)$: all other eigenvalues in $\operatorname{Re}\lambda < 0$, which is H2 together with the count of §5. That $\kappa_1(M)$ is the *first* value of κ at which the constant equilibrium loses stability is a separate statement. The verifier proved it by exact algebra for $15 \le M \le 57$ (§11): for these M, every crossing of the imaginary axis for $K > 0$ is located and the constant equilibrium is linearly stable on the mass hyperplane for all $0 < \kappa < \kappa_1(M)$. For $M > 57$, "first" is used in the sense of the paper's Theorem 3.2 (first crossing in (β_0, β_k)) and is supported only by the floating-point counts of §5.

Remarks.

- The first branch exists from $M = 15$, at $K \approx 212.9$, $\kappa \approx 824.5$. For $M \le 14$ no crossing was found for $K \le 10^5$ (floating-point scan). Branch 2 starts at $M = 44$ ($K \approx 232$) and branch 3 at $M = 87$ ($K \approx 265$). At each onset the crossing comes in from $K = \infty$. The verifier proved the first two onsets by exact algebra (§11): for $3 \le M \le 14$ the constant equilibrium is linearly stable on the mass hyperplane for every $K > 0$ and no eigenvalue ever reaches the imaginary axis; for $15 \le M \le 43$ there is exactly one crossing; for $44 \le M \le 57$ exactly two. The onset of branch 3 at $M = 87$ remains a floating-point statement.
- $M = 15$ is a genuine exception. $\ell_1 = +1.1952 \times 10^{-9}$ (unit normalization), certified, with tight radius. The dense floating-point formula gives the same value. BifurcationKit, an independent code with automatic differentiation, reports `SubCritical` there and `SuperCritical` at $M = 16, 17, 30$.

- The scale-free value ℓ_1/M (with $q_1 = 1$) is tiny at $M = 15$ ($+5.8 \times 10^{-10}$), against -6.3×10^{-6} at $M = 16$. This suggests that an analytic continuation in M would have a degenerate (Bautin) point between 15 and 16; that continuation is not computed here. For the integer problem the first branch is subcritical exactly at its first value $M = 15$.

Representative certified values. The unit column uses $q^H q = 1$; the last column is ℓ_1/M with the normalization $q_1 = 1$.

M	j	$\kappa_j(M)$	ω	ℓ_1 (unit)	ℓ_1/M ($q_1 = 1$)
15	1	824.53962734	87.797	$+1.195\times10^{-9}$	$+5.8 \times 10^{-10}$
16	1	83.095251377	8.0421	-1.296×10^{-5}	-6.27×10^{-6}
30	1	9.2400641499	0.34187	-6.723×10^{-3}	
100	1	3.9348822214	2.174e-2	-3.001×10^{-2}	
100	2	15.746601845	0.18808	-2.477×10^{-3}	
100	3	121.37739313	2.1986	-1.033×10^{-5}	
10^4	1	2.0375686900	9.36e-6	-1.542×10^{-3}	-0.011015
10^4	2	3.7981676429		-4.525×10^{-3}	-0.007971
10^4	3	5.5155380850		-3.798×10^{-3}	-0.003637
10^5	1	1.9413727095	2.778e-7	-4.028×10^{-4}	-0.009236
10^5	2	3.3593351327	1.162e-6	-1.443×10^{-3}	-0.007509
10^5	3	4.5144663668	2.465e-6	-1.347×10^{-3}	-0.003525

Full data: `certificates/sweep_j{1,2,3}*.csv.gz` (one line per M).

7 The limit $M \to \infty$

7.1 7.1 Formal limit problem

Notation: $\varepsilon = M^{-1/2}$, $K = \varepsilon\kappa$, $x = \ell/M$, $\tau = \varepsilon^3 t$ (paper (35)). Write $n_1 = 1 + c$, $c = \varepsilon\gamma$, $\gamma = \kappa + \Gamma$.

1. **Bulk.** Exactly, $\dot{n}_\ell = c\,(n_{\ell-1} - n_\ell) + (n_{\ell-1} - 2n_\ell + n_{\ell+1})$ for $2 \le \ell \le N$: upwind transport with speed c plus diffusion. With $n_\ell \approx 1 + \varepsilon\kappa + \varepsilon V(x, \tau)$, at leading order

$$V_\tau + \gamma V_x = 0 \quad (0 < x < 1), \qquad V(0, \tau) = \Gamma(\tau).$$

   The inflow value comes from $n_1 = 1 + \varepsilon\gamma$. Diffusion contributes εV_{xx}.
2. **Boundary layer at $\ell = M$**, of width ℓ-scale $1/c = \sqrt{M}/\gamma$. It is quasi-static: its relaxation time is $O(M) \ll M^{3/2}$. The flux $J_\ell = cn_\ell - (n_{\ell+1} - n_\ell)$ is constant across it and equals Kn_M at $\ell = N$. Solving the recurrence gives $n_\ell = a + C(1+c)^\ell$, where a is the outer value and $C(1 + c)^M = a(\gamma - \kappa)/\kappa$. Its excess mass is $\sum \ell(n_\ell - a) \approx M^2 \varepsilon\, a\, \Gamma/(\kappa\gamma)$.
3. **Monomers** relax on the fast time scale $1/(A(M + 2))$. Their equation is replaced by mass conservation, $\sum \ell(n_\ell - A) = 0$. With $a = 1 + O(\varepsilon)$ this gives

$$\int_0^1 xV\, dx + \frac{\Gamma}{\kappa(\kappa + \Gamma)} = 0.$$

Lagrangian clock. Let $dT = \gamma\, d\tau$. Along characteristics V carries its inflow value, so $V(x,\cdot) = G(T-x)$ with $G(T) = \Gamma$. The problem becomes the scalar *renewal equation*

$$\boxed{\frac{G(T)}{\kappa(\kappa + G(T))} + \int_0^1 s\, G(T-s)\, ds = 0} \tag{R}$$

(R) is autonomous, and its only nonlinearity is the boundary-layer term. A monotone time change preserves orbits, their stability and the side of bifurcation, so criticality can be read on (R).

Check against the paper. With $G = e^{\zeta T}$, linearizing (R) gives $\Delta(\zeta) = \kappa^{-2} + E(\zeta) = 0$, where $E(\zeta) = \int_0^1 s e^{-\zeta s} ds = (1 - e^{-\zeta}(1+\zeta))/\zeta^2$. Since $\zeta^2 e^{\zeta} \Delta(\zeta) = Q(\zeta;\kappa)$, this is exactly the paper's limit function Q (eq. (87)). Its purely imaginary roots are $\zeta = \pm i t_j$, where $\tan t_j = t_j$, $\cos t_j < 0$ and $\kappa_j^{02} = \sqrt{1+t_j^2} - 1$.

7.2 7.2 Landau coefficient of (R) (multiple scales)

Multiply (R) by κ^2:

$$G + \kappa^2 LG - G^2/\kappa + G^3/\kappa^2 + O(G^4) = 0, \qquad (LG)(T) = \int_0^1 sG(T-s)ds.$$

Expand with $\kappa = \kappa_c + \eta^2 \tilde{\kappa}$, $G = \eta G_1 + \eta^2 G_2 + \eta^3 G_3$, $G_1 = \mathcal{A}(\eta^2 T) e^{i\omega T} + \text{c.c.}$, $\omega = t_j$, $D(\zeta) := 1 + \kappa_c^2 E(\zeta)$.

- Order η: $D(i\omega) = 0$.
- Order η^2: $G_2 = \alpha \mathcal{A}^2 e^{2i\omega T} + \text{c.c.} + \beta|\mathcal{A}|^2$, with $\alpha = 1/(\kappa D(2i\omega))$ and $\beta = 2/(\kappa D(0))$.
- Order η^3: solvability of the $e^{i\omega T}$ component, using $L[\mathcal{A}(\eta^2 T) e^{i\omega T}] = e^{i\omega T}(E(i\omega)\mathcal{A} + \eta^2 E'(i\omega)\mathcal{A}')$ and $E(i\omega) = -1/\kappa^2$. Explicitly, using $\kappa^2 = \kappa_c^2 + 2\kappa_c \eta^2 \tilde{\kappa} + \ldots$, the $e^{i\omega T}$ component of the order-η^3 equation reads

$$\kappa_c^2 E'(i\omega)\mathcal{A}' + 2\kappa_c \tilde{\kappa} E(i\omega)\mathcal{A} - \frac{2}{\kappa_c}(\alpha+\beta)|\mathcal{A}|^2\mathcal{A} + \frac{3}{\kappa_c^2}|\mathcal{A}|^2\mathcal{A} = 0.$$

  Here $(G_1 G_2)_{e^{i\omega T}} = (\alpha+\beta)|\mathcal{A}|^2\mathcal{A}$ and $(G_1^3)_{e^{i\omega T}} = 3|\mathcal{A}|^2\mathcal{A}$.

The result is the amplitude equation

$$\mathcal{A}' = \sigma\tilde{\kappa}\mathcal{A} + \ell\,|\mathcal{A}|^2\mathcal{A}, \qquad \sigma = \frac{2}{\kappa^3 E'(i\omega)}, \qquad \ell = \frac{b}{\kappa^4 E'(i\omega)}, \qquad b = \frac{2}{D(2i\omega)} + \frac{4}{D(0)} - 3.$$

Exact evaluation at the Hopf points (`05_limit_coefficient.py`, C1–C3, sympy). Set $R = \sqrt{1+t^2} = 1 + \kappa^2$. The Hopf conditions give $e^{it} = -(1+it)/R$, and then:

$$E'(it) = \frac{1}{R} - i\,\frac{R+1}{tR(R-1)}, \qquad E(2it) = \frac{1-it}{2R^2}, \qquad D(0) = \frac{R+1}{2},$$

$$\operatorname{Re}\big(b\,\overline{E'(it)}\big) = -\,\frac{7R^2 - 34R + 23}{R(R+1)(5R-3)}.$$

The last identity is checked exactly in the rational parametrization $t = (s^2-1)/(2s)$, $R = (s^2+1)/(2s)$.

Signs.

- $\operatorname{Re}\sigma > 0$, since $\operatorname{Re}E'(it) = 1/R > 0$. This is the transversal crossing, consistent with paper Lemma 7.3.
- $7R^2 - 34R + 23 > 0$ iff $R > (17 + 8\sqrt{2})/7 \approx 4.0448$. Every Hopf point has $t_j \geq t_1 > 3.92$ (interval check: $\tan t - t$ is increasing on $(\pi, 3\pi/2)$ and negative at 3.92), hence $R_j > 4.0455$.
- Therefore $\operatorname{Re}\ell_j < 0$ **for every** $\boldsymbol{j}$. Equivalently, the sign is that of $-(7\kappa^4 - 20\kappa^2 - 4)$, negative since $\kappa_j^0 \geq \kappa_{\mathrm{cr}} = 1.8982 > 1.7449$.

Proposition B. Every Hopf bifurcation of the limit equation (R), at $\kappa = \kappa_j^0$, $j \geq 1$, has a negative Landau coefficient. Hence it is supercritical in the multiple-scales sense.

- The algebra is exact.
- The multiple-scales coefficient is the standard first Lyapunov coefficient, up to a positive factor; the rigorous Hopf theory for renewal equations is in Diekmann, Getto and Gyllenberg (2008).
- A direct simulation of (R) confirms the amplitude law. At $\kappa = \kappa_1^0 + \tilde{\kappa}$, a stable cycle forms with amplitude ratio (simulation/prediction) 1.038 at $\tilde{\kappa} = 0.004$ and 1.018 at 0.002 (claim C4b).

Prediction for the discrete $\boldsymbol{\ell_1}$ (heuristic matching of normalizations; confirmed numerically in §8). Normalize $q_1 = 1$, so $v_1 = 2\operatorname{Re}z_{\mathrm{Kuz}}$. Use $v_1 = \varepsilon G$, $T \approx \kappa\tau$ and $\tau = \varepsilon^3 t$. Then $c_1 = \varepsilon\kappa\ell$ and $\omega = \varepsilon^3\kappa t_j$, so

$$\frac{\ell_1(q_1 = 1)}{M} \;\longrightarrow\; L_j := \frac{\operatorname{Re}\ell_j}{t_j} = -\left.\frac{R(R-1)(7R^2 - 34R + 23)}{t\,(R+1)(5R-3)\big((R-1)^3 + R + 1\big)}\right|_{t=t_j}.$$

Values: $L_1 = -0.0093114$, $L_2 = -0.0079410$, $L_3 = -0.0037777$, $L_4 = -0.0021661$, $L_5 = -0.0013977$. Also $L_j \sim -7/(5t_j^2) \to 0^-$.

8 Convergence evidence (Observation C)

At sample points $M = 10^4, 3\cdot 10^4, \ldots, 10^8$ and $j = 1, 2, 3$, the Hopf point and ℓ_1 are certified as in §5 (`06_large_M_samples.py`):

- $\ell_1 < 0$ at every sample point;
- the relative deviation of $\ell_1(q_1 = 1)/M$ from L_j at $M = 10^8$ is 1.3×10^{-3} ($j = 1$), 3.6×10^{-3} ($j = 2$) and 6.7×10^{-3} ($j = 3$);
- the deviation decays like $M^{-0.44}$, consistent with an $O(M^{-1/2}) = O(\varepsilon)$ correction.

So the discrete coefficient appears to converge to the limit value predicted by §7. This is numerical support, not a proof, for both the formal limit and the closed form.

9 Validation of the machinery (`02_validate_l1.py`)

- **V1.** The generic Kuznetsov routine reproduces the Guckenheimer–Holmes coefficient on a planar test system to 10^{-12}.
- **V2.** Direct stiff simulation of the full system at $M = 30$, $K = K_c(1 + \delta)$. A small perturbation converges to a small stable cycle. The n_1 half-amplitude divided by the normal-form prediction $2|q_1|\sqrt{-\mu/(\omega\ell_1)}$ is $1.0096, 1.0047, 1.0024, 1.0012$ for $\delta = 4, 2, 1, 0.5\%$: the ratio tends to 1 linearly in δ.
- **V3.** BifurcationKit 0.8 (Julia) classifies the Hopf points as follows: $M = 15$ SubCritical ($\operatorname{Re}b = +1.22 \times 10^{-7}$); $M = 16$, 17 and 30 SuperCritical. Its K_c

agree with ours to 10^{-9}. This code is independent: reduced coordinates, automatic differentiation, its own normal-form code.

- **V4** (`bk_po_branch.jl`, log in `certificates/logs/bk_po_branch.log`). The branch of periodic orbits from the Hopf point is continued by orthogonal collocation (BifurcationKit).
  - $M = 16$: the branch leaves towards $K > K_c$ with amplitude$^2 \propto K - K_c$: supercritical.
  - $M = 15$: the branch leaves towards $K < K_c$ with amplitude$^2 \propto K_c - K$ (ratios 12.4 vs 12.8): subcritical. It then folds back towards $K > K_c$ at n_M-amplitude ≈ 1.2, a saddle-node of cycles, consistent with the tiny ℓ_1.

10 What is proved, what is not, and points for the verifier

Proved (computer-assisted, ball arithmetic).

- Theorem A: the sign of ℓ_1 for every $(M, j) \in \mathcal{S}$, with simplicity and transversality.
- H2 and the number of unstable pairs at $\kappa_j(M)$ for $M \le 120$ (solver), extended by the verifier to $M \le 160$ and $M = 250, 300$.
- (Verifier, exact algebra.) All crossings of the imaginary axis for $K > 0$ and $3 \le M \le 57$: none for $M \le 14$, one for $15 \le M \le 43$, two for $44 \le M \le 57$. The constant equilibrium is stable before the first one.

Exact (symbolic).

- The Landau coefficient of the limit equation (R) and its sign for every j (Proposition B, algebraic part).

Formal or not proved.

(1) *Large M is not covered rigorously.* There is no proof that $\ell_1(q_1 = 1)/M \to L_j$; §8 is numerical evidence plus certified samples. A complete answer for all $M > M_{0,k}$ needs this convergence with an explicit rate, together with Theorem A up to the resulting threshold. An ε-graded interval evaluation of the closed form is not directly possible: the inner products contain an exact cancellation of relative order ε^{-3} between the row-1 term and the bulk sum (mass conservation). A proof would have to reorganize the computation around the continuum structure of §7.

(2) *Derivation of (R)* (§7.1): formal (quasi-static boundary layer, slaved monomers, neglected diffusion). It reproduces the paper's Q at linear order. The match of §8 (to 10^{-3} at $M = 10^8$) supports it at cubic order.

(3) *Multiple-scales coefficient as the Lyapunov coefficient of (R):* standard but not re-proved here (renewal equations: sun-star calculus).

(4) *H2 for* $160 < M \le 10^5$ (except $M = 250, 300$) and the branch label j for $M > 300$. H2 is checked in floating point only for $M \le 300$ and every 50th $M \le 2000$, and not at all beyond; the label rests on continuation. The paper covers H2 for $M > M_{0,k}$ with $M_{0,k}$ not explicit. Consequently the orbital stability of the first-branch orbits is rigorous only for $16 \le M \le 160$ and conditional on H2 for $160 < M \le 10^5$.

(5) *Onset values*: $M = 15$ and $M = 44$ are now proved exactly (verifier, §11); $M = 87$ rests on floating-point scans with $K \le 10^5$. This affects only the *existence* of the branch below these values, not the certified signs.

(6) $j \ge 4$ is not certified. Floating point gives $\ell_1 < 0$ for $j \le 6$ at $M = 150$–300, and the limit coefficient is negative for every j.

Points for an independent verifier. All six points were addressed by the verifier (§11 and `verification.md`). Points (a), (b), (d) and (e) were checked; (c) was re-run on all lines with $M \le 10^4$ and on a sample beyond; for (f), two further independent codes and two formula-free tests were used.

- (a) Re-derive (2.1) and the role of H in (3.1): the choice $w_{11} \in H$.
- (b) Check the ansatz argument in §4, in particular the completeness of the family and row 1 versus the mass condition.
- (c) Re-run `04_verify_certificates.py 1 ...`. It re-proves every line from the stored midpoints alone.
- (d) Re-derive the boundary-layer mass $\Gamma/(\kappa(\kappa+\Gamma))$ and the Lagrangian reduction to (R).
- (e) Check the multiple-scales algebra (signs of c_3), using `05_limit_coefficient.py`, claims C2 and C4b.
- (f) Confirm $M = 15$ independently. Already done here by BifurcationKit's normal form (V3) and its periodic-orbit continuation (V4); a third, independent code would be welcome.

11 Independent verification

An independent verifier (Claude Opus 5.5, effort xhigh, fresh context, no access to the solver's reasoning) re-derived the key steps with its own code in `verification/`. The full report is `verification.md`. Verdict: **confirmed with corrections**. The corrections concern the scope of the stability statement, the word "first", the floating-point coverage of H2 and a step of the paper's Lemma 5.4. They are applied in §§5, 6 and 10.

- **Independent dense certificate** (`v02_dense_certify.py`). It does not use `bdlib.py`:
  - reduced coordinates $y = (v_2, \dots, v_M)$ on the mass hyperplane, so that no choice of w_{11} is needed;
  - Jacobian and bilinear form obtained from the vector field of the paper itself (exact central differences and polarisation);
  - Krawczyk operator on the full eigen-system $(\hat{A}(K) - i\omega)q = 0$, $q_k = 1$, in $2M$ real unknowns;
  - Kuznetsov's formula in ball arithmetic.

  Every integer M in $[15, 200]$ ($j = 1$), $[44, 200]$ ($j = 2$), $[87, 200]$ ($j = 3$) passed, and $M = 210, 220, \dots, 300$ on each branch: 487 points. At each point the verifier certified simplicity ($uq \neq 0$ with a nonsingular bordered system), transversality, $\omega > 0$ and the sign of ℓ_1. The enclosures agree with the solver's certificate to the double precision in which it is stored. At $M = 15$ the verifier finds $\ell_1 = +1.1951866439 \times 10^{-9}$ (unit normalisation, radius 9×10^{-45}). The cubic term was computed as a third difference of the field and vanishes. With `acb_mat.eig` on the reduced matrix, H2 and the count of unstable pairs are certified for $M \le 160$ on all branches and for $M = 250, 300$.
- **Exact algebra for small M** (`v07_first_crossing.py`, PARI/GP). Let $P(\lambda, K)$ be the characteristic polynomial of $\hat{A}(K)$ in $\mathbb{Z}[K][\lambda]$. The resultant of the even and odd parts of $P(iy, K)$ locates every $K > 0$ with a nonzero imaginary eigenvalue (rigorous real-root isolation). Exact Routh–Hurwitz counts in between then give the number of unstable eigenvalues. Results:
  - $3 \le M \le 14$: no crossing, stable for every $K > 0$;

  - $15 \le M \le 43$: exactly one crossing, at the certified K_1;
  - $44 \le M \le 57$: exactly two crossings, at K_1 and K_2;
  - the constant equilibrium is stable on $(0, K_1)$ in all these cases.

  For $M \ge 58$ the computation exceeded the memory allotted.
- **Direction of bifurcation without the Lyapunov formula.**
  - Shooting with variational equations for the small periodic orbits at fixed amplitude (`v09_po_shooting.py`): at $M = 15$ the orbits exist for $K < K_c$, with $(K - K_c)/a^2 \to -3.72 \times 10^{-5}$ (subcritical); at $M = 16$ they exist for $K > K_c$, with $(K - K_c)/a^2 = +4.06 \times 10^{-3}$ (supercritical).
  - Direct simulation at $M = 16$ and 50 (`v08_simulate.py`), for $K = K_c(1 + \delta)$, $\delta = 1$–4%. Perturbations starting inside and outside converge to one cycle. Its amplitude matches $2|q_1|\sqrt{-\mu/(\omega\ell_1)}$ to 2×10^{-4} at $M = 16$, and to 0.4–1.5% ($\propto \delta$) at $M = 50$. For $K = 0.98K_c$ the perturbations decay.

  These tests use none of the Lyapunov-coefficient machinery, so they check its sign convention end to end.
- **Large M.** A sparse $O(M)$ re-implementation (tridiagonal plus rank one in the reduced coordinates):
  - in double precision (`v03_sparse_float.py`), at 84 points (25 random $M \le 10^5$ per branch plus boundary cases): the sign agrees everywhere. The relative agreement is $\le 2 \times 10^{-7}$ for $M \le 10^4$ and $\le 9 \times 10^{-5}$ up to 10^5, the limit of double precision;
  - at 320 bits (`v04_sparse_arb.py`), at 12 points: the onsets $(15, 1)$, $(44, 2)$, $(87, 3)$ and $(16, 1)$; $M = 10^3, 10^4$; three random points; and $M = 10^5$ on each branch. κ agrees with the certificate to 10^{-40}, and ℓ_1 in both normalisations to $\le 10^{-16}$, the float64 storage limit of the certificate files.

  The Hopf point is re-located from a perturbed start in both cases. The solver's own code re-proved 1360 certificate lines (`v05_recheck_sample.py`): all $M \le 300$, 200 random $M \le 10^5$ per branch, and the boundary cases. The extra checks $\omega > 0$, $\operatorname{Im} z > 0$ and $f(\varphi) \ne 0$ or $f(\psi) \ne 0$ all pass. An audit of the six certificate files (`v01_audit_csv.py`) checks coverage, signs, $\omega > 0$ and smoothness and monotonicity in M.
- **Limit problem** (`v06_limit_algebra.py`): the identities of §7.2, the sign for every j and the values L_j are re-derived. The reduction to (R) was re-derived by hand; it remains formal.

12 Reproduction

From the project root in Git Bash:

```
outils/wsl "cd chemistry-open-problems/problems/
becker-doring-atomization-hopf/scripts && python <script>"
```

script	content	runtime
`01_check_model.py`	model identities, paper's eigenvalues and Table 1, closed form = dense	1 min

script	content	runtime
`02_validate_l1.py`	planar test, stiff simulation (M = 30), BifurcationKit cross-check	10 min
`03_certify_range.py J MMIN MMAX OUT.csv [SEED.csv]`	certified sweep (Theorem A)	12–30 ms per M
`04_verify_certificates.py STRIDE files...`	independent re-check of the sweep files	20 ms per line
`05_limit_coefficient.py`	limit problem, exact algebra, simulation of (R)	1 min
`06_large_M_samples.py`	certified samples up to $M = 10^8$, convergence to L_j	2 min
`07_spectrum_H2.py MMAX files...`	rigorous spectra, H2 and branch label for $M \leq$ MMAX	15 min
`bk_check.jl M Kc ...`	BifurcationKit normal form (independent)	2 min
`bk_po_branch.jl`	BifurcationKit periodic-orbit branch at $M = 15, 16$ (direction of bifurcation)	3 min

The verifier's scripts `v01`–`v09` are in `verification/` and are described in §11 and in `verification.md`, with their logs and run times.